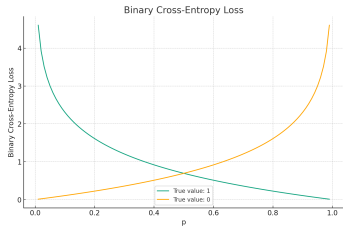
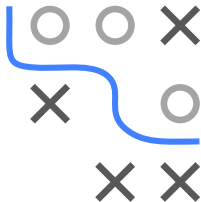


Introduction to Machine Learning

Information Theory

Cross-Entropy and KL



Learning goals

- Know the cross-entropy
- Understand the connection between entropy, cross-entropy, and KL divergence

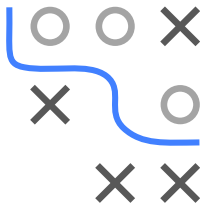
CROSS-ENTROPY - DISCRETE CASE I

Cross-entropy measures the average amount of information required to represent an event from one distribution p using a predictive scheme based on another distribution q (assume they have the same domain $\mathcal{X}$ as in KL).

$$H(p||q) = \sum_{x \in \mathcal{X}} p(x) \log \left(\frac{1}{q(x)} \right) = - \sum_{x \in \mathcal{X}} p(x) \log (q(x)) = -\mathbb{E}_{X \sim p}[\log(q(X))]$$

For now, we accept the formula as-is. More on the underlying intuition follows in the content on inf. theory for ML and sourcecoding.

- Entropy = Avg. amount of information if we optimally encode p
- Cross-Entropy = Avg. amount of information if we suboptimally encode p with q
- $DL_{KL}(p||q)$: Difference between the two
- $H(p||q)$ sometimes also denoted as $H_q(p)$ to set it apart from KL



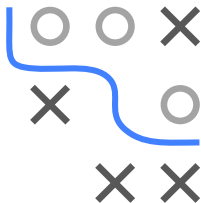
CROSS-ENTROPY - DISCRETE CASE II

We can summarize this also through this identity:

$$H(p\|q) = H(p) + D_{KL}(p\|q)$$

This is because:

$$\begin{aligned} H(p) + D_{KL}(p\|q) &= - \sum_{x \in \mathcal{X}} p(x) \log p(x) + \sum_{x \in \mathcal{X}} p(x) \log \frac{p(x)}{q(x)} \\ &= \sum_{x \in \mathcal{X}} p(x) (-\log p(x) + \log p(x) - \log q(x)) \\ &= - \sum_{x \in \mathcal{X}} p(x) \log q(x) = H(p\|q) \end{aligned}$$

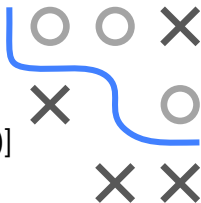


CROSS-ENTROPY - CONTINUOUS CASE I

For continuous density functions $p(x)$ and $q(x)$:

$$H(p\|q) = \int p(x) \log \left(\frac{1}{q(x)} \right) dx = - \int p(x) \log (q(x)) dx = -\mathbb{E}_{X \sim p}[\log(q(X))]$$

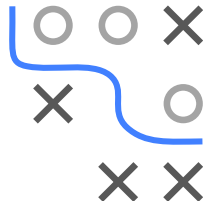
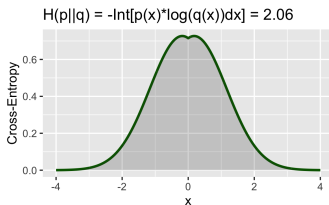
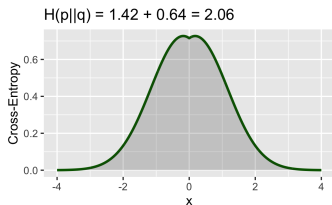
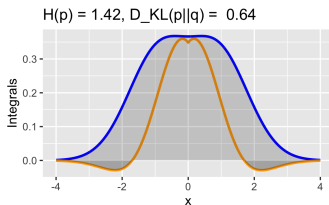
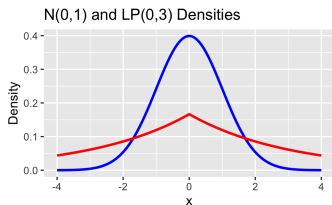
- It is not symmetric.
- As for the discrete case, $H(p\|q) = h(p) + D_{KL}(p\|q)$ holds.
- Can now become negative, as the $h(p)$ can be negative!



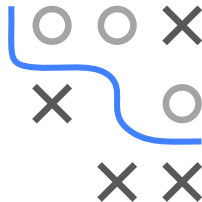
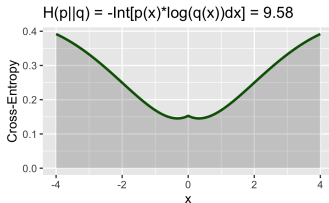
CROSS-ENTROPY EXAMPLE I

Let $p(x) = N(0, 1)$ and $q(x) = LP(0, 3)$. We can visualize

$$H(p||q) = H(p) + D_{KL}(p||q)$$



Let $p(x) = LP(0, 3)$ and $q(x) = N(0, 1)$. We can visualize

$$H(p||q) = H(p) + D_{KL}(p||q)$$


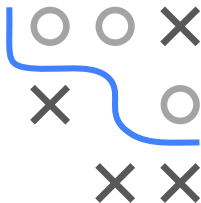
PROOF: MAXIMUM OF DIFFERENTIAL ENTROPY I

Claim: For a given variance, the continuous distribution that maximizes differential entropy is the Gaussian.

Proof: Let $g(x)$ be a Gaussian with mean μ and variance σ^2 and $f(x)$ an arbitrary density function with the same variance. Since differential entropy is translation invariant, we can assume $f(x)$ and $g(x)$ have the same mean.

The KL divergence (which is non-negative) between $f(x)$ and $g(x)$ is:

$$\begin{aligned} 0 \leq D_{KL}(f||g) &= -h(f) + H(f||g) \\ &= -h(f) - \int_{-\infty}^{\infty} f(x) \log(g(x)) dx \end{aligned} \quad (1)$$



PROOF: MAXIMUM OF DIFFERENTIAL ENTROPY II

The second term in (1) is,

$$\begin{aligned}\int_{-\infty}^{\infty} f(x) \log(g(x)) dx &= \int_{-\infty}^{\infty} f(x) \log \left(\frac{1}{\sqrt{2\pi\sigma^2}} e^{-\frac{(x-\mu)^2}{2\sigma^2}} \right) dx \\&= \int_{-\infty}^{\infty} f(x) \log \left(\frac{1}{\sqrt{2\pi\sigma^2}} \right) dx + \log(e) \int_{-\infty}^{\infty} f(x) \left(-\frac{(x-\mu)^2}{2\sigma^2} \right) dx \\&= -\frac{1}{2} \log(2\pi\sigma^2) - \log(e) \frac{\sigma^2}{2\sigma^2} = -\frac{1}{2} (\log(2\pi\sigma^2) + \log(e)) \\&= -\frac{1}{2} \log(2\pi e\sigma^2) = -h(g),\end{aligned}\tag{2}$$

where the last equality follows from the normal distribution example of the entropy chapter. Combining (1) and (2) results in

$$h(g) - h(f) \geq 0$$

with equality when $f(x) = g(x)$ (following from the properties of Kullback-Leibler divergence).

